

INTERNATIONAL SCHOOL

Methodology of Natural Science
and Technology

Introduction to Research

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Q1: Have you ever joined a research team?
 Q2: Have you ever joined in any research project?

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LEARNING OBJECTIVES

- 1) Understand what the research is
- 2) Choose the research topic
- 3) Write the research objective

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What is Research?



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What is Research?

- **“Research”** is a common term to answer any questions, results of research are all around us

“Research is the process of finding solutions to a problem after a thorough study and analysis of the situational factors”



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Why do the research?

- Get a degree
- Get scholarships or awards
- Get a job
- Learn new skills and knowledge
- Service to society

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Research's Characteristics

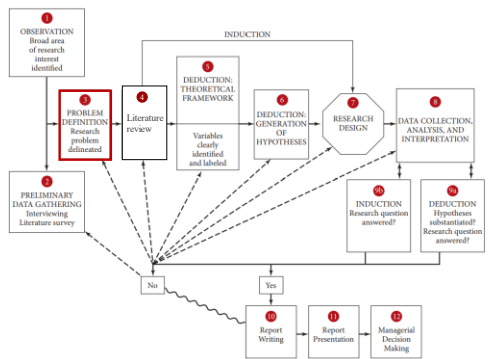
Endless	✓	Boring	✓
Never perfect	✓	Time-consuming	✓
Findings	✓	Unpredictable	✓
Public	✓	Costly	✓

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RESEARCH WORKFLOW

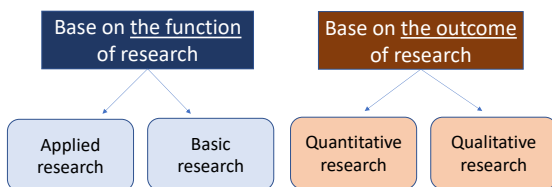


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Types of research



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Types of research

➤ **Basic research:** to generate a body of knowledge by trying to comprehend how certain problems occur



➤ **Applied research:** to solve a current problem faced, demanding a timely solution



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Basic and Applied Research

Basic research

Purpose:

- Expand knowledge of the process
- Results in universal principles relating to the process and its relationship to outcomes
- Findings of significance and value to society in general

Context:

- Undertake by people based in universities
- Choice of topic and objectives determined by the researcher
- Flexible time scale

Applied research

Purpose:

- Results in solution to the problem
- Findings of practical relevance and value to manager(s) in the organization(s)

Context:

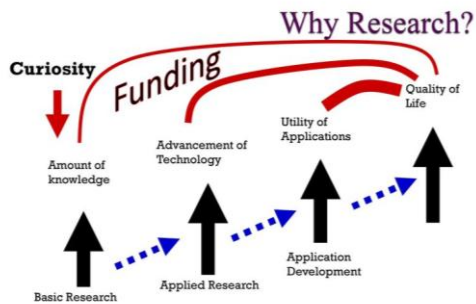
- Undertake by people based in a variety of settings including organizations and universities
- Objective negotiated with the originator
- Tight time scale

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Types of research



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Types of research

➤ **Quantitative research:** Measured by size, quantity, percent, frequency, cost, etc.



➤ **Qualitative research:** Identify, explore, and understand the nature, the characteristic and cause of the problem



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Types of research

➤ **Quantitative research:**

Eg:

- ✓What is the average weight of Vietnamese women?
- ✓How is the effectiveness of robot arms in surgery?

➤ **Qualitative research:**

Eg:

- ✓Why do people increasingly prefer to watch short videos instead of long movies?
- ✓What factors influence university students' decision to choose a major of study?

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Choose topic and Research questions

Practice:

Choose 01 research topic. Make 04 questions corresponding to 04 types of research: Basic research, Applied research, Quantitative research, and Qualitative research.

Eg:

- **Topic:** The obesity of Vietnamese people
- **Basic research:** What is the mechanism of obesity?
- **Applied research:** How is the effectiveness of drug A to lose weight of obese people in Vietnam?
- **Quantitative research:** What percentage of obese people in Vietnam?
- **Qualitative research:** Why are Vietnamese people afraid to get the drug to lose weight?

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Choose the research topic

- Health: The average weight of Vietnamese women, ...
- Tech: Robot arms in surgery, ...
- Social: Watch short videos, ...
- Education: University students' decision to choose a major of study

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Choose the research topic



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Choose the research topic

Topic	Give a score from 1-3						Total score	Order of priority
	Urgency	Applicability	Possibility	Level of impact	Community concern	Ethical acceptance		
1								
2								
3								

Name of topic:
1.
2.
3.

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Research objective

Practice: Research on the frequency of using artificial intelligence tools by students at the International School in 2024.

- Objective 1: ...
- Objective 2: ...

Summary question

- 1) What is the research?
- 2) Why do we need to do research?
- 3) How many types of research?
- 4) What are the main factors in choosing the research topic?
- 5) How to write the name of a research topic?
- 6) How to write the research objective?

Take-home message

“Science is not belief, but the will to find out.”

**THANK YOU
FOR THE KIND ATTENTION!**

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